COMPOUND INTEREST CALCULATOR

AIM:

To implement a COBOL program to perform compound interest calculation using the provided principal amount, rate percentage and number of years.

ALGORITHM:

Step 1: Start the program.
Step 2: Define the identification division details and the required variables such as PRINCIPAL, RATE, YEARS, TOTAL and PRINT-TOTAL along with their input value format.
Step 3: In procedure division accept the inputs and check whether they satisfy the range specified. If not display "INVALID".
Step 4: Then, move PRINCIPAL to TOTAL, perform YEARS times the interest calculation for a year and move TOTAL to PRINT-TOTAL to get the correct formatted output without any leading zeros and rounded to two decimal places.
Step 5: Display the output and STOP RUN.

PROGRAM:

```
IDENTIFICATION DIVISION.
PROGRAM-ID. COMPOUND_INTREST.
AUTHOR. JOHN.

DATA DIVISION.
    WORKING-STORAGE SECTION.
        01 PRINCIPAL PIC 9(7).
        01 RATE PIC 9(2).
        01 YEARS PIC 9(2).
        01 TOTAL PIC 9(14)V99.
        01 PRINT-TOTAL PIC Z(13)9.99.

PROCEDURE DIVISION.
    ACCEPT PRINCIPAL
    ACCEPT RATE
    ACCEPT YEARS
```

```
    IF PRINCIPAL >1000000 OR YEARS > 50 OR RATE > 30
        DISPLAY "INVALID"
    ELSE
        MOVE PRINCIPAL TO TOTAL

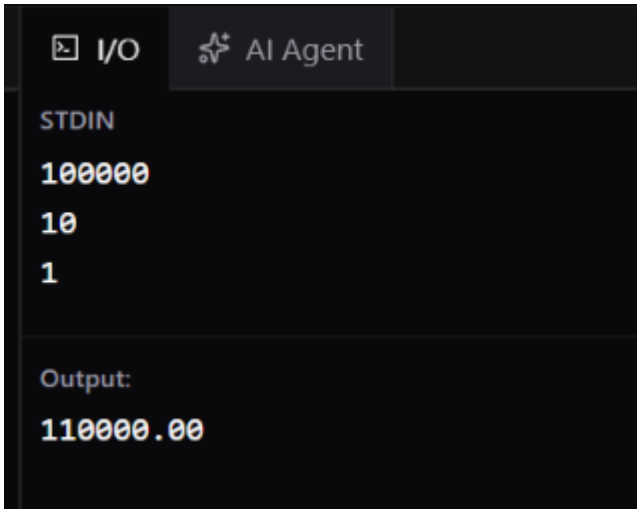
      PERFORM YEARS TIMES
        COMPUTE TOTAL ROUNDED = TOTAL * (1 + RATE/100)
      END-PERFORM

      MOVE TOTAL TO PRINT-TOTAL
      DISPLAY FUNCTION TRIM(PRINT-TOTAL)

    END-IF

STOP RUN.
```

OUTPUT:



RESULT:

Thus, the COBOL program to calculate compound interest is successfully executed and the outputs are verified.